AI Engineer Assesment Task by Newera.ai

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Role	AI Engineer
Project	Newera HR Assistant (Conversational HR Self-Service)
Time Budget	7th June, 2026
Primary Language	**Arabic-first** (English optional)
Delivery	Source repository that runs with **one command via Docker**

1. Context

Newera builds enterprise AI assistants. This assignment evaluates your ability to design and ship a complete **agentic, RAG-backed HR chatbot** end-to-end — from data modeling and tool design to retrieval accuracy, frontend UX, and security.

You will build a chatbot that allows a Newera employee to log in and ask natural-language questions about their own HR data and company policies. The assistant must autonomously decide which capability or tool to use for each query.

This is a **build assignment**. We prioritize working software, thoughtful engineering decisions, and excellence across the four evaluation areas detailed in Section 7.

2. What You Will Build

A secure web application where a logged-in employee can ask questions in **Arabic** (English is a bonus) across five HR domains. The assistant intelligently routes each query to the appropriate capability and provides answers grounded in real data or official policy documents.

2.1. Query Domains

For domains (a–d), design and populate a realistic mock database with seeded data for 2-3 employees. For domain (e), implement a robust **RAG pipeline** over the provided Newera employee handbook.

2.2. Agent Behavior

- The LLM must intelligently select the correct capability/tool for each query — **no manual mode switching** by the user.
- Tools should preferably be exposed via **MCP (Model Context Protocol)**. The agent acts as the MCP client.
- The assistant must support **write operations** (e.g., submitting leave requests and logging overtime) and provide clear confirmation to the user.
- Maintaining **chat history** across turns within a session is a strong plus.

#	Domain	Example Questions	Data Source
a	Attendance	كم يوم تأخرت هذا الشهر؟ أظهر أوقات تسجيل الدخول الأسبوع الماضي.	Mock DB
b	Leave Requests	كم يوم إجازة سنوية متبقية لي؟ قدم طلب إجازة لمدة 3 أيام بدءاً من الأحد.	Mock DB
c	Salary	ما هو صافي راتبي الشهر الماضي؟ فصّل كشف الراتب الأخير.	Mock DB
d	Overtime	كم ساعة عمل إضافي لدي هذا الشهر؟ سجّل ساعتين عمل إضافي ليوم أمس.	Mock DB
e	Policy	كم يوم مرضي أنا مستحق؟ ما هي سياسة العمل عن بعد؟	RAG (Newera Handbook)

3. Technical Constraints (Hard Requirements)

You are free to choose your stack, but the following are mandatory:

1. **Local LLM only.** Use a model runnable on a typical laptop (∼4B parameters recommended, e.g., via Ollama, llama.cpp, or vLLM). **No cloud LLM APIs.** Clearly state the model and runtime used.
2. **MCP Tools.** HR capabilities must be exposed as MCP tools (including RAG tool).
3. **RAG for Policy.** All policy answers must be grounded in retrieved handbook content (not parametric memory). Retrieval quality is a primary evaluation metric.
4. **Arabic-first.** Full support for Arabic input, output, and RTL UI rendering.
5. **One-command Docker deployment.** The full system should start with `docker compose up` (LLM serving may be a documented prerequisite).

4. Frontend Requirements

- Secure login flow with session-based authentication.
- **Newera visual identity**: Reference newera.ai for colors, typography, and modern enterprise feel.
- Full Arabic-first UI with proper RTL layout, fonts (e.g., Noto Sans Arabic), and locale-aware formatting.
- Clean, professional chat interface with message history, loading states, error handling, and optional tool-routing visibility.

5. Security Requirements

Security is explicitly graded. Minimum expectations:

1. Strict cross-user data isolation — enforced server-side.
2. Robust prompt injection resistance (tested against malicious inputs).
3. Secure authentication (hashed passwords, protected sessions).

4. Input sanitization and protection against SQL injection.

Document your threat model and mitigations in `DESIGN.md` or the README.

6. Deliverables

Submit a Git repository containing:

1. Complete source code.
2. Docker Compose setup for one-command execution.
3. Comprehensive `README.md` with setup instructions, model used, test credentials, and example Arabic queries.
4. Auto-seeded mock database (minimum 3 employees).
5. Design documentation (`DESIGN.md`) covering architecture, agent design, RAG pipeline, and security.

7. Evaluation Rubric

#	Area	Weight	Evaluation Criteria
1	UI / UX	25%	Newera-themed, polished Arabic-first interface with excellent RTL support and intuitive flows.
2	RAG Accuracy	30%	Faithful, hallucination-free policy answers grounded in the handbook.
3	Code Quality & Engineering	25%	Clean architecture, thoughtful tool/agent design, maintainable code, and solid Docker setup.
4	Security	20%	Strong isolation, injection resistance, and documented threat model.

Bonuses: Persistent chat history, bilingual support, tool call transparency, automated tests, and thoughtful UX for write operations.

8. Ground Rules & Tips

- Seed realistic data for **at least 3 employees**.
- Prioritize correctness, security, and accuracy over feature breadth.
- Keep secrets out of the repository (use `.env.example`).
- Provide clear, incremental commit history.

9. Submission

Share the repository link by the agreed deadline. A reviewer must be able to clone, run, and test the application using only your documentation.

Good luck! We look forward to seeing your solution.

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